

Chaeyeon Yoo

chaeyeon.yoo@uconn.edu

Phillip E. Austin Building, Storrs, CT 06269-4120

EDUCATION

University of Connecticut

Graduate (Advisor: Dipak K. Dey)

Ph.D in Statistics

Aug.2020 - Present
Connecticut, United States

Ewha Woman University

Graduate: Master (Advisor: Jae Keun Yoo)

Master of Science in Statistics(Applied Statistics)

Mar. 2018 - Feb. 2020
Seoul, Korea

Ewha Woman University

Undergraduate

Bachelor of Science in Statistics and Minor in Economics

Mar. 2014 - Feb. 2018
Seoul, Korea

PUBLICATION

Publication in progress

Yoo C, Dey DK., Im Y *Bayesian variable selection with correlation spike and slab prior for cure rate model*

Yoo C, Dey DK. *Gaussian Process-Augmented Promotion Time Cure Models with Bayesian Variable Selection for Incidence Modeling.*

Publication under review

Yoo C, Lachos Davila VH, Carrasco JMF, Dey DK. *Mixture cure models with flexible survival and link functions: Parameter estimation via the EM algorithm (under review).*

R Package for flexible mixture cure models via EM algorithm

Yoo C, Dey DK. *Predictive Modeling of Short-Term Recidivism: A Mixture Cure Rate Approach with Diverse Link Functions (under review).*

Publication accepted

Yoo C, Yoo Y, Um HY, Yoo JK. *On hierarchical clustering in sufficient dimension reduction. Communications for Statistical Applications and Methods (CSAM).* 2020;27:431–443.

<https://doi.org/10.29220/CSAM.2020.27.4.431>

COLLABORATIVE PUBLICATION

Collaborative Research

Kim D, Defino A, Eason C, Casa D, Yoo C, Yoshihara A, Coleman K, Filep E, Huggins R. *Athletic Training Services Trends Between Public and Private High Schools: A 5-Year Retrospective Analysis.* (Accepted)

Kim D, Grubb J, Eason C, Casa D, Yoo C, Yoshihara A, Coleman K, Filep E, Huggins R. *Evolution of Athletic Training Services Trends in U.S. High Schools with Athletics: A 5-Year Retrospective Analysis.* (submitted)

Srinivasan, S. M., Kataria, K., Yorns, W., Jr., Moore, T. E., Yoo, C., Kumavor, P. D., ... Friel, K. (2025). *A novel program including ride-on toys to improve upper extremity function in children with hemiplegia: a randomized controlled trial.* Disability and Rehabilitation: Assistive Technology, 1–14.

<https://doi.org/10.1080/17483107.2025.2534436>

RESEARCH & CONSULTING

Research Assistant for Travelers Project (Advisor: Dipak K. Dey) Dec. 2022 - Present
Department of Statistics, University of Connecticut Connecticut, United States
Investigation on Demand Surge Induced by Natural Catastrophic Events

- Demand Surge on reconstruction costs, induced by natural catastrophic perils are analyzed.

Development on the Uplift Model for the Application to Customer Services

- Application of optimal dynamic treatment rule (ODTR) estimation to marketing.

Consultant Sep. 2023 - Present

Statistical Consulting Service at University of Connecticut

- Analysis on effect of adipocyte conditioned medium on lipid composition of two cancer cells
- Analysis on the effect of a mindfulness-based affirmative program for LGBTQIA+ youth
- Simulation on interrater agreement with intraclass coefficient (ICC) and overall concordance correlation coefficient (OCCC) for small sample size
- Analysis on dental student survey on silver diamine fluoride application
- Identify genes with significant difference in expression between the treatment and control groups
- Propensity score calculation for determining the effect of summer camps on school attendance
- Sample Size Calculation for testing the effectiveness of a supplement on oral mucositis duration
- Randomization application for subject allocations
- Examine the relationship between perceived benefits and barriers to physical activity among Black women to inform future interventions.
- Analysis on the effectiveness of physical therapy intervention on children
- Examining sex differences on heat acclimation

Consultant Sep. 2019 - Feb.2020

Ewha Institute of Statistics, Ewha Womans University

- Conducted practical analysis requested by clients in the field of Education of Art History

Consult Assistant Sep. 2018 - Aug. 2019

Ewha Institute of Statistics, Ewha Womans University

TEACHING EXPERIENCE

Teaching Assistant Aug.2020 - May.2023
Department of Statistics, University of Connecticut Connecticut, United States

Introduction to Statistics I

Elementary Concepts of Statistics

Introduction to Mathematical Statistics I

Teaching Assistant Mar. 2018 - Feb. 2020
Department of Statistics, Ewha Womans University Seoul, Korea

Mathematical Statistics I

Mathematical Statistics II

Bayesian Statistics and Statistical Computing

Regression Analysis

Tutoring for R programming Dec. 2019
Suwon Hi-tech High School Gyeonggi-do, Korea

HONORS AND AWARDS

Awards

Student Poster Award, ICSA 2025 Applied Statistics Symposium

Champion for UCONN Campus, 2021 Travelers Statistical Modeling Competition

Poster Award, 2019 Fall Conference of the Korean Statistical Society

Entry in Big Data Championship, Statistics Department of Korea, 2019

Scholarships and Fellowships

CLAS Summer Fellowship for the Summer, University of Connecticut, 2025
Pre-doctoral Fellowship for Spring, University of Connecticut, 2025
Conference Participation Award from The Graduate School, University of Connecticut, 2024
CLAS Summer Fellowship for the Summer, University of Connecticut, 2024
CLAS Summer Fellowship for the Summer, University of Connecticut, 2023
Research Assistant Scholarships, Ewha Womans University, 2018
Admission Scholarships, Graduate, Ewha Womans University, 2018
Honors Scholarships, Ewha Womans University, 2016

Grant Proposal

Contributed to the proposal writing for an SPF Grant titled "*New Statistical Tools to Analyze Multidimensional and Multiple Sources of Data with Applications*". (Rejected)

Contributed to the proposal writing for an R21 Grant for submission to *Disability and Rehabilitation: Assistive Technology*. (Rejected)

REFERENCES

Dipak K. Dey

Department of Statistics, University of Connecticut
Email: dipak.dey@uconn.edu
Phone: +1 (860) 486-4755

Victor Hugo Lachos Dávila

Department of Statistics, University of Connecticut
Email: hlachos@uconn.edu
Phone: +1 (860) 486-3413

Timothy Moore

Department of Statistics, University of Connecticut
Email: timothy.e.moore@uconn.edu
Phone: +1 (860) 486-0074